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Everett Gorin

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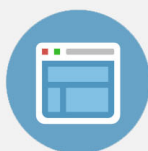
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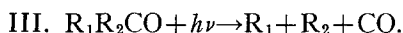
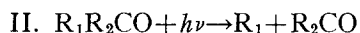
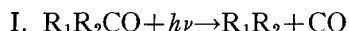
EVERETT GORIN*

Frick Chemical Laboratory, Princeton, New Jersey

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The photolysis of acetone and of formaldehyde takes place principally through a free radical mechanism. The primary processes consist in the splitting off of a methyl radical and a hydrogen atom, respectively. The primary process in the photolysis of methyl ethyl ketone also consists in the splitting off of either a methyl or an ethyl radical. The heat of dissociation of the first hydrogen in formaldehyde is shown to be as low as 78 kcal. Approximately 26 kcal. is required to remove the second hydrogen atom.

A PRELIMINARY report of this investigation has been published recently.¹ In addition a paper² on the photolysis of acetaldehyde has been presented for publication. Leighton³ in a recent review article on the photolysis of aldehydes and ketones remarks that one of the present problems in the photochemistry of aldehydes and ketones is the determination of the relative importance of the three primary reactions



The purpose of the present investigation was to determine the relative probabilities of these three reactions for the simpler aldehydes and ketones. The method consisted in the quenching of the free radicals primarily formed by iodine vapor to form the corresponding alkyl halides. Saturated hydrocarbons that may be formed by direct splitting from the aldehyde or ketone do not react with iodine in the temperature range investigated. By an analysis of the reaction products, it is thus possible to distinguish quantitatively, primary reactions giving free radicals from those yielding saturated hydrocarbons. The R_2CO radicals were found to be unusually stable, enabling one to distinguish between reactions II and III.

I. ACETONE

The experimental procedure was almost identically the same as that already described in the case of acetaldehyde.² Acetone of C.P. grade was purified by recrystallization with sodium iodide. It was further twice fractionally distilled in vacuum at 10°C before admitting into the reaction vessel. The vapor was slowly mixed with iodine vapor, to avoid condensation of the iodine and resultant formation of solid iodo compounds with acetone. In the vapor phase, blank experiments performed up to 150°C, showed no detectable thermal reaction with the iodine. As previously, the quantum yield of the reaction with iodine was determined by comparison with the direct photolysis at 60°C. This was nearly the temperature employed in the experiments of Damon and Daniels⁴ on the quantum yields of the acetone photolysis. Their value of 0.17 at high intensities was adopted. (At low intensities they give a value of 0.29 ± 0.06 at 3130 and 0.33 at 2650Å. They give no measurements at high intensities at 2650Å but we have assumed that since the yields are equal at these two wavelengths for low intensities that it would also be true for high intensities. At any rate this assumption turned out to be self consistent in that the quantum yields measured in this way were nearly unity for both filtered and unfiltered light.) The quantum yield increases at low intensities but none of our measurements were carried out at intensities low enough for this effect to be important. The methods of analysis

* This work was carried out at the Karpov Physico-Chemical Institute in Moscow U.S.S.R.

¹ *Acta Physicochimica* 8, 513 (1938).

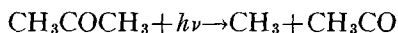
² *Acta Physicochimica* 9, 681 (1938).

³ P. A. Leighton, *J. Phys. Chem.* 42, 749 (1938).

⁴ G. H. Damon and F. Daniels, *J. Am. Chem. Soc.* 55, 2363 (1933).

for CH_3I and CO were the same as described previously in the case of acetaldehyde.²

Since the quantum yield, according to Damon and Daniels, is independent of the wave-length, the quantum yield of the reaction with iodine could be determined by direct comparison of the rates in the full light of a mercury arc. The results of these experiments are given in Table I. The first column gives the pressure of acetone vapor reduced to 60°C, the second the initial pressure of iodine in mm, the third the temperature, the fourth and fifth the number of moles times 10^6 of CH_3I and CO formed, the sixth the time of illumination and the seventh gives the quantum yield for the formation of CH_3I . The quantum yield for the formation of CH_3I is thus unity, so that the primary process must be simply



with an efficiency of unity. This conclusion is essentially in agreement with that of Taylor and Rosenblum,⁵ who concluded on the basis of experiments on the photolysis in the presence of hydrogen that acetone decomposes principally through a free radical mechanism.

It is also seen that, in the presence of iodine and at temperatures below 80°, only a very slight amount of carbon monoxide is formed. This points to a high stability of the acetyl radical. The stability of the acetyl radical has in fact been demonstrated earlier by the detection of biacetyl among the products of the photolysis at room temperature.^{6, 7}

Acetyl iodide was determined in the reaction products by distilling in vacuum over a heated mercury trap to remove the excess iodine and

TABLE II. *Experiments in unfiltered light on the stability of the acetyl radical and the formation of acetyl iodide.*

PRESS. ACETONE	INIT. PRESS. IODINE	MOLES $\times 10^6$ CH_3I	MOLES $\times 10^6$ CH_3COI	MOLES $\times 10^6$ CO	TEMP. °C
165	4	21.7	9.3	1.62	100
487	2	43.0	4.0	2.67	100
430	4	29.4	8.8	5.28	115
156	2	36.8	4.6	5.82	115
517	4	14.6	7.0	4.96	128
169	2	41.6	3.4	15.02	130

then collecting in a liquid-air trap. The resultant mixture was freed of methyl iodide by distilling off this substance with 80 percent of the acetone at -30°C in vacuum. The residual liquid was treated with water to hydrolyze the acetyl iodide forming hydriodic acid. An analysis for hydriodic acid by oxidation with bromine water to iodate followed by addition of excess iodide to liberate free iodine was then carried out. The results of this analysis are shown in Table II. It is seen that the reaction is incomplete, only from 10–40 percent of the acetyl radicals combining with iodine. The fact that the amount of acetyl iodide formed appears to be roughly proportional to the iodine pressure, would suggest that the incompleteness of this reaction is not due to slowness but to the rapid establishment of the equilibrium.



If the rate of recombination of iodine atoms were given by $-d\text{I}/dt = k_1(\text{I})^2P$, where P is the total pressure, and of acetyl radicals to form biacetyl by $-d(\text{CH}_3\text{CO})/dt = k_2(\text{CH}_3\text{CO})^2$ then, under the assumption of the above equilibrium, the amount of acetyl iodide formed would be simply proportional to the iodine concentration at the end of the reaction and independent of the extent of the reaction

$$(\text{CH}_3\text{COI})_{\text{final}} = K \left(\frac{k_1 P}{k_2} \right)^{\frac{1}{2}} (\text{I}_2)_{\text{final}}$$

K is the equilibrium constant for reaction I. It is seen that there is qualitative agreement between this equation and the experimental results.

No attempt was made to analyze quantitatively for diacetyl, but its presence was demon-

TABLE I. *Experiments in unfiltered light.*

EXP.	PRESS. MM ACETONE	PRESS. MM IODINE	TEMP. °C	MOLES $\times 10^6$ CH_3I	MOLES $\times 10^6$ CO	TIME MIN.	$\gamma_{\text{CH}_3\text{I}}$	MOLES $\times 10^6$ CH_3COI
1	652	4	80	21.4	0.40	30	1.00	—
2	652	0	60	—	3.63	30	—	—
3	365	2	80	17.6	0.39	30	0.91	—
4	365	0	60	—	3.29	30	—	—
5	616	1	80	18.7	0.36	30	0.93	2.5
6	616	0	60	—	3.42	30	—	—
7	561	1	60	19.4	0.10	30	0.94	2.2
8	561	0	60	—	3.51	30	—	—

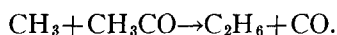
⁵ H. S. Taylor and C. Rosenblum, *J. Chem. Phys.* **6**, 119 (1938).

⁶ M. Barak and D. W. G. Style, *Nature* **135**, 307 (1935).

⁷ R. Spence and W. Wild, *Nature* **138**, 206 (1936); *J. Chem. Soc.* 352 (1937).

strated qualitatively by treating the residual liquid with hydroxylamine hydrochloride and then precipitating the red nickel dimethyl glyoxime salt.

Spence and Wild found that, although a considerable quantity of biacetyl was formed at room temperature, upon raising the temperature to 60°C no trace of biacetyl could be found. From our results, however, it is evident that this is not due to decomposition of the acetyl radical into methyl and carbon monoxide but to the secondary reaction.



As a matter of fact, from the amount of carbon monoxide formed (in the presence of I_2) it is evident that even at 100°C only about eight percent of the acetyl radicals decompose before combining to give either acetyl iodide or biacetyl. The amount of carbon monoxide increases with the temperature, however, as shown in Table II. From the temperature coefficient of the carbon monoxide formation, a rough *minimum* value of 17 kilocalories for the energy of activation of the decomposition into methyl radicals and carbon monoxide is deducible.

The results in filtered light of wave-length 3130A are given in Table III. The quantum yield for the formation of methyl iodide is slightly less than unity. This may be due to some inactivation of the excited acetone molecules by collision, since the presence of fluorescence^{3, 8} points to a rather long-lived excited molecule. Damon and Daniels³ give an upper limit of three percent for loss of absorbed energy due to fluorescence. Matheson and Noyes⁹ have recently shown that the fluorescent spectra of acetone and biacetyl are identical for wave-lengths longer than 5000A, indicating that acetone itself is not the emitting species. These facts taken together show that the loss of energy due to fluorescence cannot cause an appreciable lowering of the quantum efficiency.

Spence and Wild⁷ have proposed, from the fact that the ratio $\text{C}_2\text{H}_6/\text{CO}$ decreases towards unity in filtered light of wave-length 3130A, that,

⁸ R. G. W. Norrish, H. G. Crone and O. D. Saltmarsh, J. Chem. Soc. 1456 (1934).

⁹ M. S. Matheson and W. A. Noyes, J. Am. Chem. Soc. 60, 1857 (1938).

TABLE III. Experiments in filtered light. $\lambda = 3130\text{A}$.

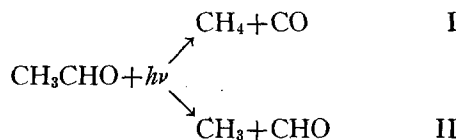
EXP.	PRESS. ACETONE	PRESS. IODINE MM	TEMP. °C	MOLES $\times 10^6$ CH_3I	MOLES $\times 10^6$ CO	TIME MIN.	$\gamma_{\text{CH}_3\text{I}}$
1	852	1	90	5.67	—	120	0.85
2	852	0	80	—	1.13	120	—
3	1265	0.5	90	8.76	—	120	0.78
4	1265	0	80	—	1.90	120	—

at these wave-lengths, the predominant process might be the direct splitting off of C_2H_6 . Our results do not support this conclusion and the primary process at 3130A must remain predominantly $\text{C}_2\text{H}_6\text{CO} + h\nu \rightarrow \text{CH}_3 + \text{CH}_3\text{CO}$.

Damon and Daniels⁴ report that the quantum yield increases at low intensities. It is difficult to understand this, as, on the basis of the proposed mechanism, there should be no dependence of the quantum yield on the light intensity. The only possibility is that a reaction of the methyl radicals with acetone becomes important at low intensities. However, this reaction if it does occur, cannot set up reaction chains, since Leermakers¹⁰ found that the quantum yield for the photolysis is only slightly greater than unity even at 400°C.

II. ACETALDEHYDE

This case has been treated in detail elsewhere² so that we will content ourselves with a brief recapitulation of the main conclusions reached. Two primary reactions were found to take place



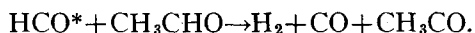
The sum of the quantum yields for both reactions was found to be unity within the experimental error at both wave-lengths studied, $\lambda = 3130\text{A}$ and $\lambda = 2536\text{A}$. At 3130A reaction II is 2.6 times as probable as reaction I, while at 2536A reaction I is 2.9 times as probable as reaction II. The low quantum yield at $\lambda = 3130\text{A}$ must thus be due to recombination of the free radicals, methyl and formyl, as in the case of acetone. The increased probability of reaction I at lower wave-lengths thus explains the increase

¹⁰ J. A. Leermakers, J. Am. Chem. Soc. 56, 1899 (1934).

in the quantum yield from 0.3 to 0.9 observed by Leighton and Blacet.¹¹

The formyl radicals were found to be very stable with respect to the decomposition $\text{HCO} \rightarrow \text{H} + \text{CO}$ and reacted mainly to give formaldehyde and CO i.e., $\text{HCO} + \text{HCO} \rightarrow \text{H}_2\text{CO} + \text{CO}$. At temperatures above 100°C the reaction $\text{HCO} + \text{I}_2 \rightarrow \text{HI} + \text{I} + \text{CO}$ is observed. The temperature coefficient of this reaction gives a minimum value of 26 kcal. for the activation energy for the reaction $\text{HCO} \rightarrow \text{H} + \text{CO}$.

Blacet and Roof¹² have observed an increase in the amount of hydrogen produced at the lower wave-lengths. This has been interpreted¹³ as meaning that the free radical mechanism becomes more predominant at 2537Å than at 3130Å. This is in disagreement with our results. The production of hydrogen at 2537Å, however, may possibly be explained by a direct reaction of an energy rich formyl radical with acetaldehyde.



No polymerization was found in the presence of iodine, indicating that polymerization is induced by free radicals. The decrease in the amount of polymerization with wave-length again confirms the conclusion that the free radical mechanism becomes less probable at low wave-lengths.

According to Leighton and Blacet¹¹ the quantum efficiency of the straight photolysis at 3130Å increases with increasing intensity. This suggests that, as in the case of acetone, secondary reactions of the free radicals with acetaldehyde take place even at low temperatures. Blacet and Volman,¹⁴ have in fact, recently suggested, to account for this fact, a chain mechanism for the photolysis even at 30°C. The quantum efficiency for the primary process, however, remains unity under all conditions.

III. FORMALDEHYDE

Paraformaldehyde was purified by warming in a stream of dried nitrogen and freezing out in a liquid-air trap. The liquid air was then removed

from the trap and the escaping monomeric formaldehyde was repolymerized by impinging upon a cold surface. A weighed amount of this polymer was then introduced into the apparatus. The bulb in which the formaldehyde was placed was first evacuated thoroughly and then, after closing the valve to the pumps, it was heated with an auxiliary furnace to 160°C. The resulting monomeric formaldehyde was then allowed to diffuse slowly, through a magnetically operated glass valve, into the reaction bulb containing a known amount of iodine vapor. The pressure of formaldehyde was calculated from a knowledge of the volume of the system. The quantum yield for the reaction in the presence of iodine was determined, as before, by comparison with the yields for the decomposition in the absence of iodine as reported by Norrish and Kirkbride.¹⁵ They report a quantum yield of 0.7 at 3650Å and very close to unity at lower wave-lengths.

The isolation of the lines in the mercury arc around 2500Å and around 3100Å was carried out as before.² In addition experiments were made at 3650Å. This line was isolated by a glass filter containing a layer 1 cm in depth of a 0.03 percent solution of diamantfuxin.

After illumination, the remaining free iodine was removed by passing the formaldehyde gas through a trap containing a pool of heated mercury and into a liquid-air trap. The permanent gases were pumped off and analyzed for CO by oxidation over I_2O_5 and for H_2 by combustion with O_2 over a heated platinum wire as described previously.² Hydrogen iodide was analyzed by removing the liquid air and allowing the unpolymerized formaldehyde along with the hydrogen iodide to distill into a trap containing a concentrated solution of potassium hydroxide. Part of the formaldehyde, however, invariably polymerized and occluded hydrogen iodide in it. This part was accordingly dissolved in water and added to the part which distilled over into the KOH. It was necessary to introduce a slight modification in the previous method for the analysis of iodide² since the presence of formaldehyde interfered with the procedure. After acidifying, the iodide was accordingly precipitated as silver iodide and collected and washed

¹¹ P. A. Leighton and F. E. Blacet, J. Am. Chem. Soc. **55**, 1766 (1933).

¹² F. E. Blacet and J. G. Roof, J. Am. Chem. Soc. **58**, 278 (1936).

¹³ G. K. Rollefson, J. Phys. Chem. **41**, 259 (1937).

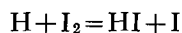
¹⁴ F. E. Blacet and D. Volman, J. Am. Chem. Soc. **60**, 1243 (1938).

¹⁵ R. G. W. Norrish and F. W. Kirkbride, J. Chem. Soc. **42**, 1518 (1932).

on a glass filter. The silver iodide was then oxidized to silver bromide and iodate ion by passing successive portions of bromine water through the filter. After removing the excess bromine by steam distillation the iodate was treated with iodide, liberating iodine, which was titrated with standard thiosulfate.

The results are given in Table IV. The first column gives the wave-length, the second the pressure of formaldehyde, the third the pressure of iodine, the fourth, fifth and sixth columns give the number of moles of HI, CO and H₂ times 10⁶, respectively. The experiments with and without iodine were always done in pairs so that the seventh column gives the ratio of the rates of formation of carbon monoxide for the experiments in the presence of iodine to the successive experiments in the absence of iodine. The next to the last column gives the ratio of the amount of HI formed to that of CO. The last column gives the quantum yield for the formation of HI.

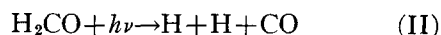
It is seen that at 3130A and 2537A the permanent gases consist almost exclusively of carbon monoxide, there being only a few percent of hydrogen formed. In addition, the HI/CO ratio is nearly equal to two for all the experiments at these wave-lengths. These facts show that the main reaction is splitting off hydrogen atoms. Since it takes 102 kcal. to pull off the two hydrogen atoms from formaldehyde it is obvious that a quantum of light of wave-length 3130A possesses only sufficient energy to detach one hydrogen atom. The primary process at 3130A must therefore be $\text{H}_2\text{CO} + h\nu \rightarrow \text{H} + \text{HCO}$ (I). This, in the presence of iodine, would be followed by



and $\text{HCO} + \text{HCO} = \text{H}_2\text{CO} + \text{CO}$.

We see that the quantum efficiency for the formation of hydrogen iodide is about unity, and that the rate of formation of CO in the presence of I₂ is about $\frac{1}{2}$ as large as it is in the absence of iodine, as is required by the above mechanism.

At 2537A the primary process



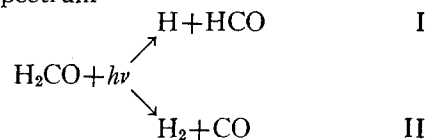
becomes energetically possible. If this were the case, the quantum efficiency for the formation of HI should become equal to two while

TABLE IV. Formaldehyde + I₂ at 100°C.

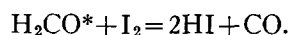
WAVE-LENGTH A	PRESS. H ₂ CO MM	PRESS. I ₂ MM	MOLES ×10 ⁶ HI	MOLES ×10 ⁶ CO	MOLES ×10 ⁶ H ₂	$d[\text{CO}]/\text{I}_2$	HI/CO	γ_{HI}
						$d[\text{CO}]$	CO	
3130	85	4	12.6	7.0	0.1	0.53	1.8	0.97
3130	88	0	—	13.0	12.5	—	—	—
3130	115	2	17.4	9.2	0.1	0.51	1.9	0.96
3130	117	0	—	18.1	17.7	—	—	—
3130	215	1	19.3	11.6	0.1	0.61	1.7	0.84
3130	212	0	—	16.2	15.8	—	—	—
2537	96	2	11.3	6.5	0.0	0.57	1.7	0.95
2537	96	0	—	11.8	11.9	—	—	—
2537	157	5	21.7	13.0	0.1	0.55	1.7	0.86
2537	154	0	—	25.2	25.0	—	—	—
2537	246	5	8.6	5.2	0.1	0.51	1.65	0.85
2537	247	0	—	10.1	9.9	—	—	—
3650	154	3	14.6	13.6	6.3	H_2/CO 0.46	HI/H_2 2.4	—
3650	152	0	—	15.2	15.4	1.00	—	—
3650	240	1	9.7	9.1	4.4	0.48	2.2	—
3650	239	0	—	12.2	11.6	0.95	—	—
Exp. with HgI ₂ vapor as acceptor								
3650	175	4	14.8	12.5	5.3	0.42	2.8	—
3650	177	0	—	14.9	14.1	0.95	—	—

$d[\text{CO}]_{\text{I}_2}/d[\text{CO}]$ should now be unity. There is no significant change in the values at 2537A as compared to those at 3130A, so that we may conclude that the primary process remains principally I. A simultaneous occurrence of II to the extent of ten percent is not, however, precluded by our data.

At 3650A a considerable quantity of hydrogen is formed. However hydrogen iodide is formed also and more carbon monoxide is formed than hydrogen. The ratio is about two as was previously the ratio HI/CO. This points therefore to two independent primary processes. The ratio of the probabilities of these two reactions is equal to the HI/H₂ ratio and is as 0.7 is to 0.3. That is, if the sum of the quantum yields of the two reactions is unity, that of I is 0.7 and of II is 0.3. The spectrum



of formaldehyde¹⁶ in this wave-length region exhibits fine structure and consequently the excited molecule must have a relatively long life. It might be objected therefore that we are dealing with a direct reaction between the excited formaldehyde molecule and iodine



This is very unlikely since the quantum yield for

¹⁶G. H. Dieke and G. B. Kistiakowsky, Proc. Nat. Acad. Sci. 18, 367 (1932).

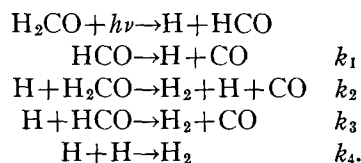
the formation of HI is independent of the iodine pressure. In order to show this even more conclusively HgI_2 vapor was substituted for I_2 as an acceptor for H atoms. The ratio of the probabilities for these two reactions was found to be the same as with I_2 .

These results lead to a surprisingly small value for the heat of dissociation of the first hydrogen in formaldehyde. A mole quantum of light of wave-length 3650A possesses energy equivalent to 78 kilocalories, consequently the heat of dissociation of the first hydrogen is not greater than 78 kcal. Mecke¹⁷ has identified the predissociation limit in the absorption spectrum with the first dissociation and consequently assigned to it a value of 102 kcal. The predissociation limit, however, can only give an upper limit^{18, 19} to the heat of dissociation. Thus in SO_2 ²⁰ the predissociation limit corresponds to an energy of 13 kcal. greater than the heat of dissociation into SO and O.

Patat²¹ has found that in the photochemical oxidation of formaldehyde the ratio of H_2/CO is greater than unity in the continuum while it is very nearly equal to unity in the banded region. He concluded from this fact that hydrogen atoms are primarily formed in the continuum but not in the banded region. Carruthers and Norrish²² have shown however that these experiments cannot be used to detect hydrogen atoms since formic acid is the first product of the oxidation, a large portion of the permanent gases being derived from the subsequent photolysis of the formic acid. The reaction was also found to be a chain reaction with a large quantum yield.

Akeroyd and Norrish²³ in studying the chain photolysis of formaldehyde have used a suggestion made previously²⁴ that the primary process consists in the rapid successive breaking off of the two hydrogen atoms. Our results show,

however, that only one hydrogen atom is ejected in the primary process and that, furthermore, the formyl radical is quite stable, about 26 kilocalories being required to dissociate it. This stability of the formyl radical has also been remarked upon by earlier workers.^{25, 26} We therefore propose to replace Akeroyd and Norrish's scheme for the chain photolysis by the following one



This leads to the equation for the steady-state concentration of hydrogen atoms.

$$I_{\text{abs}} - \frac{k_4}{2}(\text{H})^2 - \frac{k_3 k_4}{2k_1}(\text{H})^3 = 0. \quad (1)$$

Now, k_1 has a large temperature coefficient corresponding to 26 kilocalories and therefore, at low temperatures, we should expect the second term to be dominant while at high temperatures the third term will be the largest. Thus, in the lower temperature range, the rate should be given approximately by

$$-\frac{d(\text{H}_2\text{CO})}{dt} = I_{\text{abs}} + k_2 \left(\frac{2I_{\text{abs}}}{k_4} \right)^{\frac{1}{2}} (\text{H}_2\text{CO}), \quad (2)$$

while at higher temperatures the rate would be given by

$$-\frac{d(\text{H}_2\text{CO})}{dt} = I_{\text{abs}} + k_2 \left(\frac{2k_1 I_{\text{abs}}}{k_3 k_4} \right)^{\frac{1}{2}} (\text{H}_2\text{CO}). \quad (3)$$

The temperature coefficient at the higher temperatures thus should be considerably greater than at the lower temperatures. This is just what Akeroyd and Norrish found, there being a rather sharp change in the temperature coefficient between 200 and 250°C. Now k_3 should have a small activation energy. If we use five kilocalories for k_3 and the value 26 kcal. found for k_1 then, using the experimental value of 16.2 kcal. for the over-all activation energy of the photolysis at high temperatures, we find a value of 9.2 calories for k_2 . This is of the order of magnitude of the

¹⁷ R. Mecke, *Zeits. f. Electrochemie* **36**, 589 (1930).

¹⁸ G. Herzberg, *Zeits. f. Physik* **61**, 604 (1930); *Zeits. f. physik. Chemie* **B10**, 189 (1930).

¹⁹ L. A. Turner, *Phys. Rev.* **68**, 178 (1931).

²⁰ J. Franck, H. Sponer and E. Teller, *Zeits. f. physik. Chemie* **B18**, 88 (1932).

²¹ F. Patat, *Zeits. f. physik. Chemie* **B25**, 208. F. Patat and T. Locker, *Zeits. f. physik. Chemie* **B27**, 431 (1935).

²² J. E. Carruthers and R. G. W. Norrish, *J. Chem. Soc.* 1036 (1936).

²³ E. I. Akeroyd and R. G. W. Norrish, *J. Chem. Soc.* 890 (1936).

²⁴ C. H. Bamford and R. G. W. Norrish, *J. Chem. Soc.* 1504 (1935).

²⁵ F. Patat, *Zeits. f. physik. Chemie* **B32**, 274 (1936).

²⁶ M. Burton, *J. Am. Chem. Soc.* **60**, 212 (1938).

value 10 kcal. obtained for the similar reaction involving acetaldehyde.²⁵ Part of the fall in the temperature coefficient must of course be attributed to the constant terms in (2) and (3). The data are not sufficient to correct accurately for this effect, but a rough estimate of the temperature coefficient in the low temperature region, taking this into account gives a value of 6-8 kcal., in fair agreement with the value for k_2 calculated from the high temperature region.

METHYL ETHYL KETONE

Some experiments were also carried out with methyl ethyl ketone in the presence of I_2 . Since the quantum yield in the near ultraviolet apparently has not been previously determined, no comparison of the rates in the presence and absence of I_2 was made. A few comparisons were made, however, of the rates of the reactions of iodine with acetone and with methyl ethyl ketone under comparable conditions. In addition to CO some hydrocarbon was formed. The carbon hydrogen ratio of the hydrocarbon was determined in much the same way as previously was done for methane.² The analyses for alkyl iodide and carbon monoxide were carried out in the same way as previously. Experiments were carried out both in filtered and unfiltered light. The results are given in Table V.

The gaseous products were collected by means of a high speed diffusion pump after freezing out the ketone in liquid air. The vapor pressure of propane is of the order of 1.5×10^{-3} mm at liquid-air temperatures, but since the diffusion pump pumped down to 10^{-6} mm with a fore vacuum of 0.05 mm, it was fairly certain that all the propane was collected. The pressure of gas collected in the fore-vacuum of the diffusion pump was of the order of 0.05 to 0.1 mm. At any rate, after reducing the fore vacuum to 10^{-5} mm and thawing out and refreezing the ketone, no appreciable quantity of gas was further collected. The analysis of the carbon hydrogen ratio C_nH_m was not accurate due to the small quantities available for analysis. It corresponded to the value for propane, however, within the limits of experimental error.

We see from the results in Table II that methyl ethyl ketone behaves very much like acetone. The amount of CO formed is very small com-

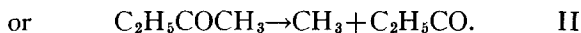
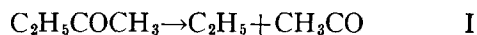
TABLE V. *The photolysis of methyl ethyl ketone in the presence of iodine at 70°C.*

PRESS. MeCOEt MM	WAVE- LENGTH	PRESS. IODINE MM	MOLES $\times 10^6$ RI	MOLES $\times 10^6$ CO	MOLES $\times 10^6$ C_3H_8
65	3130	4	22.2	1.05	.82
70	3130	2	21.2	.89	.74
67	Unfiltered	4	17.2	1.24	1.12
72	Unfiltered	2	17.6	1.33	1.24

Comparison of rates of photolysis of acetone and methyl ethyl ketone in the presence of iodine and unfiltered light.

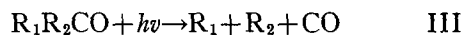
PRESS.	SUBSTANCE	TIME	MOLES $\times 10^6$ RI
90	MeCOEt	40	24.6
90	Acetone	40	21.7
90	MeCOEt	40	22.1
90	Acetone	40	20.6

pared with the amounts of alkyl iodide indicating that the principal primary reaction is either

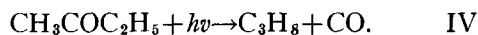


Since no attempt was made to analyze separately for CH_3I and C_2H_5I our results are incapable of distinguishing between the two. As before diketones were detected qualitatively by precipitation as the nickel glyoxime salt. The fact that Norrish and Appleyard²⁷ found that ethane, propane and butane were formed in equal amounts in the direct photolysis would seem to indicate that I and II have about equal probabilities.

It is to be noted that the amount of CO always exceeds the amount of C_3H_8 . This is probably to be attributed to a slight thermal decomposition of the acyl radicals. However a few percent of the type of reaction postulated by Norrish^{24, 27} and co-workers cannot be definitely



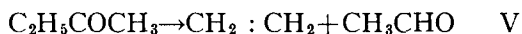
ruled out. The formation of propane must be explained by the independent primary reaction



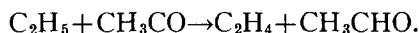
In unfiltered light about 6.5 percent of the ketone decomposes in this way while in filtered light of wave-length 3130A 3.5 percent decomposes in this way.

²⁷ R. G. W. Norrish and M. E. S. Appleyard, J. Chem. Soc. 874 (1934).

Norrish and Appleyard²⁷ postulate for the higher ketones a new kind of primary process represented by the reaction



in the case of methyl ethyl ketone. Such an over-all reaction could also possibly be explained by a secondary reaction such as



If ethylene diiodide were found among the products of the reaction with iodine then it would be good evidence that reaction V occurs in a single step. However, no attempt was made in the present work to detect $\text{C}_2\text{H}_4\text{I}_2$.

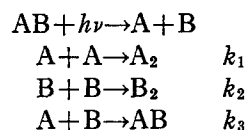
In the last three experiments we have compared the rates of the reactions of iodine with MeCOEt and Me_2CO . We see that the MeCOEt reaction is about ten percent faster than the reaction with acetone under the same conditions. The absorption coefficient of MeCOEt in the liquid state²⁸ averages about 20 percent higher than that of acetone while in alcohol²⁹ it is about five percent higher. It is probable that it is, therefore, also slightly higher in the vapor phase. We may conclude therefore, that the quantum yield for the primary process in MeCOEt is also about unity and that reaction V postulated by Norrish and Appleyard²⁷ most likely occurs by secondary processes.*

RECOMBINATION OF FREE RADICALS

It is a common characteristic of the photolysis of aldehydes and ketones that though the quantum yield for the primary process is very nearly unity, the net yield at low temperatures is only a fraction of this. As has been shown above, this low yield is due to the recombination of the free radicals primarily produced. The increase in the yield at higher temperatures is due to two factors. The first is the setting in of a chain reaction and the second an increasing probability for the reaction $\text{R}_1 + \text{R}_2\text{CO} \rightarrow \text{R}_1\text{R}_2 + \text{CO}$ in place of their simple recombination. Even in the case of formaldehyde, though the quantum yield is about unity at 100° the lowest temperature measured,

the yield of the nonchain part of the reaction is less than unity. This can be shown by fitting the kinetic expression $d(\text{HCHO})/dt = k_1 + k_2(\text{HCHO})$ to the data of Akeroyd and Norrish²⁸ at 100°C. The constant k , corresponds to the nonchain part of the reaction and since it is responsible for about $\frac{1}{2}$ of the over-all reaction at this temperature, the quantum yield for the nonchain part is only about 0.5.

It is easy to show that if we had a system of reactions of the type



and that even if the activation energies for all three reactions were zero, that the quantum yield would be only about $\frac{1}{3}$.

If there were no activation energy in the true sense of the word, the two radicals would be attracted to one another by polarization forces. This attractive force will be opposed by the centrifugal force of the rotation of the two free radicals considered as a diatomic rotator. There will thus be a maximum of potential energy at quite a large distance of separation, whose position will depend upon the amount of rotational energy, and which corresponds to the activated complex. The vibrational and rotational degrees of freedom of the free radicals will be relatively unaffected due to their large separation in the activated complex. The problem of their rate of recombination can be treated then as if they were atoms and follows exactly the treatment given for a similar case by Eyring, Hirschfelder and Taylor.³⁰ This calculation has been given in more detail in (2) and we will simply write down the result for the rate constant

$$k = 3(2)^{1/6} J\left(\frac{2}{3}\right) \left(\frac{\alpha_A \alpha_B I_A I_B}{I_A + I_B} \right)^{1/2} \times \frac{\pi^{1/2} (kT)^{1/6}}{\sigma_{AB}} \frac{M_A M_B}{(M_A + M_B)^{5/2}}.$$

α_A and α_B are the polarizabilities of radicals A and B, I_A and I_B are the ionization potentials,

²⁸ Beilecki and Henri, Ber. **45**, 2819 (1912).

²⁹ Beilecki and Henri, Ber. **46**, 3627 (1913).

* See, however, C. H. Bamford and R. G. W. Norrish, J. Chem. Soc. 1531 (1938).

³⁰ H. Eyring, J. O. Hirschfelder and H. S. Taylor, J. Chem. Phys. **4**, 479 (1936).

M_A and M_B the masses, and $J(2/3)$ the gamma-function of two-thirds. σ_{AB} is the symmetry number for the two free radicals considered as the diatomic molecule. Now for k_1 and k_2 , $\sigma=2$ while for k_3 , $\sigma=1$. The rest of this expression does not show much variation for the three reactions. Therefore $k_1 : k_2 : k_3 \sim 1 : 1 : 2$ and the quantum yield.

$$\gamma = \frac{1}{1 + k_3(1/k_1 k_2)^{1/2}} = \frac{1}{1 + 2} = \frac{1}{3}.$$

It should be remarked here, that on this basis Norrish and Appleyard²⁷ should have found a ratio 1 : 4 : 1 of ethane to propane to butane as against their observed value of 1 : 1 : 1, in the products of the photolysis of methyl ethyl ketone, provided that the two free radicals ethyl and methyl were split off simultaneously in the primary process. Our results show, however, that only one radical is split off at a time and that therefore the formation of the hydrocarbons proceeds mainly through reactions of the type $\text{CH}_3 + \text{C}_2\text{H}_5\text{CO} \rightarrow \text{C}_3\text{H}_8 + \text{CO}$ and $\text{CH}_3 + \text{CH}_3\text{CO} \rightarrow \text{C}_2\text{H}_6 + \text{CO}$ etc. These reactions no doubt would involve small activation energies so that it would be impossible to predict the relative amounts of ethane propane and butane that would be formed. If the formation of a methyl or an ethyl radical in the primary process had equal probability and the above secondary reactions had equal rates then the ratio of ethane to propane to butane should be 1 : 2 : 1.

BOND ENERGIES

From thermochemical data³¹ the reaction $\text{HCHO} \rightarrow 2\text{H} + \text{CO}$ is endothermic to the extent of 104 kcal. We have derived above a maximum value of 78 kcal. for the strength of the first bond in formaldehyde and a value of 26 kcal. for the strength of the second bond. It is thus seen that in order for their sum to be equal to 104 kcal. we must identify them with their real values. Previ-

ously it had been assumed, on the basis of identifying the predissociation limit in formaldehyde with the strength of the first bond, that nearly all of the 104 kcal. was required to dissociate the first hydrogen and that the second dissociated with great ease.

It is also possible with these values, and the value of 17 kcal. found for the reaction $\text{CH}_3\text{CO} \rightarrow \text{CH}_3 + \text{CO}$ to deduce a value for the strength of the first bond in methane not inconsistent with other estimates. We assume that it also takes 78 kcal. to remove the aldehyde hydrogen in acetaldehyde. This, of course, is not strictly true, but we should expect that the C-H in acetaldehyde would be within five kcal. of that in formaldehyde. This value combined with the value 17 kcal. for the dissociation of the acetyl radical and the fact that the reaction $\text{CH}_3\text{CHO} \rightarrow \text{CH}_4 + \text{CO}$ is exothermic to extent of one kcal. gives 96 kcal. for the heat of dissociation of the first hydrogen in methane.

Trenner, Morikawa and Taylor³² have derived a value of 108 kcal., from the heats of activation that they were able to obtain for the reactions $\text{CH}_4 + \text{D}$ and $\text{CH}_3 + \text{HD}$. Polanyi and Hartel³³ have deduced an upper limit of 110 kcal. from the fact that chlorine atoms react rapidly with methane. They also give a lower limit of 98 kcal. derived from observations on the reactions $\text{CH}_3 + \text{H}_2$ and $\text{CH}_4 + \text{H}$. Because of an uncertainty of possibly as much as five kcal. in thermochemical data for the reaction $\text{CH}_3\text{CHO} \rightarrow \text{CH}_4 + \text{CO}$ it is possible to bring our value within these limits. It does not seem possible, however, accepting the value 78 kcal. for the heat of dissociation of the first hydrogen in acetaldehyde to obtain a value quite as high as Trenner's.

I wish to express my appreciation to Professors A. Frumkin and A. Rabinowitsch for the hospitality extended to me during my stay at the Karpov Institute.

³² N. R. Trenner, K. Morikawa and H. S. Taylor, *J. Chem. Phys.* 5, 203 (1937).

³³ H. v. Hartel and M. Polanyi, *Zeits. f. physik. Chemie* 11, 97 (1931).

³¹ F. R. Bichowsky and F. D. Rossini, *Thermo Chemistry of Chemical Substances* (1936).